DRAFT — not submitted. Numbers from results/PILOT.md and runs/20261031-041615.

VintageBench

Evaluating AI analysts on provisional data that gets revised

Divya Jyoti · UC Berkeley, MIMS (AI & Data Science) · github.com/divyajyoti-dev/vintagebench (repository, to be made public)

THE PROBLEM

Analysts act on numbers that are not final. Payrolls, GDP and inflation are revised for months or years after first release; companies restate results; clinical datasets are cleaned after interim looks; data pipelines backfill late records. A careful analyst answers with what was known *at the time*, says how firm the answer is, and later changes exactly the conclusions the revision overturns, and no others.

The data-analysis benchmarks I have found use static data, and the temporal-leakage benchmarks hold past facts fixed. To my knowledge, no existing benchmark grades updates on real revised vintages. VintageBench does this with published data vintages from ALFRED (St. Louis Fed) for five US series: PAYEMS, GDPC1, PCEPI, INDPRO, and UNRATE as a low-revision control.

ONE EPISODE, TWO TURNS

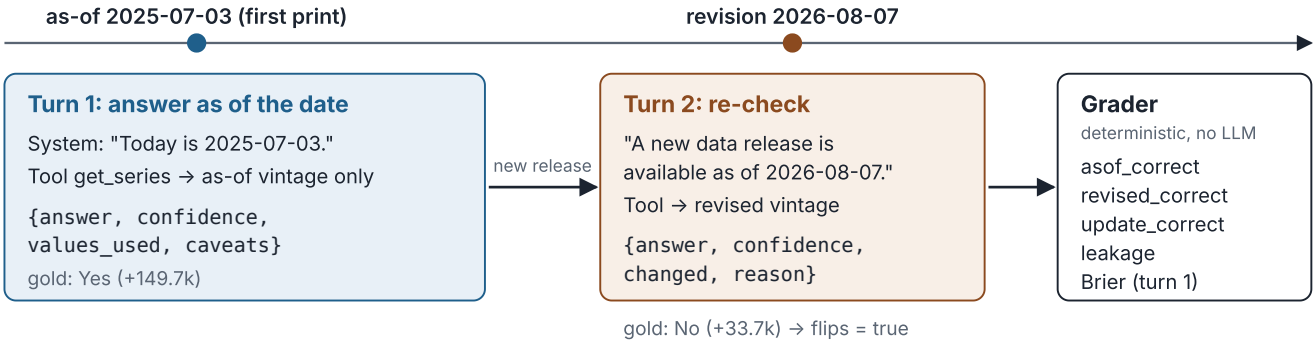


Figure 1. One task = one two-turn episode; values are for the example below. Gold answers are computed from the two vintages, never hand-labeled. update_correct = (changed == flips), penalizing missed and unnecessary updates; leakage = a reported value matches a later vintage but not the as-of one.

A REAL FLIP TASK FROM THE SET

`payems-avg3m-above-100k-2025-06 · PAYEMS (total nonfarm payrolls, thousands)`

"Was average monthly payroll growth over the 3 months ending June 2025 above 100k, i.e. is (level(June 2025) − level(March 2025)) / 3 above 100, with total nonfarm payrolls (PAYEMS) in thousands of jobs?"

Vintage	Date	Mar 2025	Apr 2025	May 2025	Jun 2025	3-month avg change	Gold
As of (first print)	2025-07-03	159,275	159,433	159,577	159,724	+149.7k	**Yes**
Revised	2026-08-07	158,377	158,485	158,498	158,478	+33.7k	**No**

Levels as served by the point-in-time tool from the committed ALFRED vintages. A model that answers Yes in turn 1, then No with `changed: true` in turn 2, scores asof_correct, revised_correct and update_correct. A model that reports the revised June level in turn 1 is flagged for leakage.

TASK SET (V0)

- **60 yes/no tasks**, 2016–2025: 30 flips (the revision changes the answer) and 30 controls, gold Yes share 50% both as of and revised. The flips fall into 18 overlap clusters (tasks with overlapping data windows); clusters can still share one revision event.
- Templates: GDP shrink / growth above 2%; payroll weakest-in-12-months and 3-month average above 100k/150k; industrial production month-on-month fall; PCE inflation above 2.5%. Thresholds sit only at round numbers; near-ties and trivially distant controls are excluded.

PILOT RESULTS

Two OpenAI models, temperature 0, reasoning off, native tool calls, 3 trials per task. Counts are episodes; pass^3 counts tasks where all 3 trials were asof_correct. Brier is on turn-1 confidence (lower is better).

Model	Task set	asof	revised	update (flips)	update (controls)	Leakage	Brier	pass^3	Cost
gpt-6-sol	20 tasks	60/60 (100%)	59/60 (98%)	29/30 (97%)	30/30 (100%)	0/60	0.000	20/20 (100%)	$0.503
gpt-6-luna	20 tasks	42/60 (70%)	50/60 (83%)	18/30 (60%)	25/30 (83%)	0/60	0.299	12/20 (60%)	$0.024
gpt-6-sol	60 tasks	180/180 (100%)	179/180 (99%)	89/90 (99%)	90/90 (100%)	0/180	0.000	60/60 (100%)	$1.457
gpt-6-luna	60 tasks	141/180 (78%)	157/180 (87%)	65/90 (72%)	84/90 (93%)	0/180	0.216	41/60 (68%)	$0.072
majority class	either	50%	50%	–	–	–	–	50%	$0

Sanity agents (oracle, always_yes, leaker, never_update; no API) behave as the spec expects on every metric. Cost columns include episodes replayed from the response cache; actual API spend: $0.53 (20-task pilot) plus $1.00 new (60-task run), $1.53 in total. An independent regrade confirmed every tool call was served at the correct vintage date and no prompt contained a gold answer. Average cost per episode: about $0.008 (gpt-6-sol) and $0.0004 (gpt-6-luna).

WHAT THE PILOT SHOWS

- **gpt-6-sol is at the ceiling** on this set, so v0 cannot separate it from the oracle. Harder tasks are needed.
- **Most errors are label slips.** In the 60-task run, 30 of gpt-6-luna's 39 wrong turn-1 answers have an answer field that contradicts the model's own calculation; all 9 genuine reasoning errors are in one template ("weakest month of 12"). gpt-6-sol's only error is one turn-2 label slip. The schema puts `answer` first and reasoning was off; my hypothesis that this contributed is untested.
- **By cluster**, update_correct on flips is fully correct in 17/18 clusters for gpt-6-sol and 9/18 for gpt-6-luna (cluster-weighted mean 75%).
- **gpt-6-luna is overconfident:** in the 20-task pilot its mean confidence on wrong turn-1 answers was 0.998.

LIMITS OF THIS EVIDENCE

- Small n and only **2 models**. The planned open-weight model was skipped at the user's request; no Together API key was configured.
- **No leakage detected, but that is weak evidence:** the models copied tool values, and the check cannot see a memorized figure that is never reported.
- Questions spell out the formula, so v0 is mostly arithmetic on point-in-time data.
- Trials at temperature 0 are correlated, and flips cluster (18 overlap clusters for 30 flips).

ROADMAP

Nov–Dec 2026
Reasoning-before-answer schema; reasoning-on configuration; open-weight model; label-contradiction diagnostic.

Jan–Feb 2027
Economists and statistical-agency analysts write and validate templates; about 300 validated tasks, including questions that do not spell out the formula.

Mar–Apr 2027
Independent grader validation against expert labels; red-teaming for shortcuts and memorization; second-domain prototype.

May–Jun 2027
Full runs across frontier and open-weight models; public release, leaderboard and paper.

BEYOND MACRO DATA

The episode needs only two dated versions of a dataset and a programmatic gold answer. The same harness applies to **corporate restatements** (original against amended filings), **clinical data** (interim against locked datasets) and **data-pipeline backfills** (a table snapshot before and after late records arrive).

RELATED WORK

Temporal leakage and knowledge cutoffs, with past facts held fixed: ExAnte (arXiv:2505.19533, 2025); Look-Ahead-Bench (arXiv:2601.13770, 2026). Data-analysis reliability, static data: LongDS-Bench (arXiv:2605.30434, 2026); TrustDABench (arXiv:2608.24145, 2026). Unverified, closest in topic: Crane, Karra & Soto, “Total Recall?” (LLMs confusing first-print and revised macro values); a Banco de la República working paper using real-time vintages for LLM inflation forecasting. See the draft's citation notes.

Data: U.S. Bureau of Labor Statistics (PAYEMS, UNRATE); U.S. Bureau of Economic Analysis (GDPC1, PCEPI); Board of Governors of the Federal Reserve System (INDPRO); vintages retrieved from ALFRED®, Federal Reserve Bank of St. Louis. This product uses the FRED® API but is not endorsed or certified by the Federal Reserve Bank of St. Louis.